

Math 237B Midterm

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1 D

Problem 1. Let $A = \mathbb{C}[x, y]/(f(x, y))$ and consider the scheme $X = \text{Spec}(A)$. Prove or disprove that there is a finite set $S \subset X$ such that every connected component of $X \setminus S$ is irreducible as a topological space.

Solution. We'll prove there exists a finite set $S \subset X$, such that all connected component of $X \setminus S$ is irreducible as topological space.

First, let's cover some special cases like $f = 0$ or $f = a$ for some nonzero $a \in \mathbb{C}$:

- If $f = 0$, then $A = \mathbb{C}[x, y]$ (an integral domain), as a result $X = \text{Spec}(A)$ is irreducible, hence connected. Take $S = \emptyset$, then $X \setminus S = X$ has itself being a connected component, which is irreducible.
- If $f = a$ ($a \neq 0$ in $\mathbb{C}$), then $A = 0$ as an abelian group (since $(f(x, y)) = \mathbb{C}[x, y]$). So, $\text{Spec}(A) = \emptyset$ is a trivial case we can ignore.

Hence now on, we'll assume f is a non-constant polynomial.

Given that $A = \mathbb{C}[x, y]/(f(x, y))$, one knows that $X = \text{Spec}(A) \cong V((f)) \subset \text{Spec}(\mathbb{C}[x, y])$ as subspace topology. Hence, we'll mainly work with this subspace.

The proof will follow this order:

- (1) Show that $V((f)) = \bigcup_i V(P_i)$ for finitely many distinct prime ideals P_i .
- (2) Show that $S := \bigcup_{(i_1, \dots, i_k)} \left(\bigcap_{j=1}^k V(P_{i_j}) \right)$, i.e. the union of any intersection of distinct $V(P_i)$'s, is a finite set, and moreover only contains maximal ideals.
- (3) Show that $X \setminus S = \left(\bigcup_i V(P_i) \right) \setminus S = \bigsqcup_i (V(P_i) \setminus S)$, and each nonempty $V(P_i) \setminus S$ is irreducible (hence connected).
- (4) Deduce that each nonempty $V(P_i) \setminus S$ must be the connected component of $X \setminus S$, hence $X \setminus S$ has all connected components being irreducible as topological spaces.

Proof of (1):

First, recall that $\mathbb{C}[x, y]$ is a UFD, hence $f(x, y) = \prod_{i=1}^n (f_i(x, y))^{k_i}$ for finitely many non-associate irreducible polynomials $f_1, \dots, f_n \in \mathbb{C}[x, y]$, and some powers $k_1, \dots, k_n \in \mathbb{N} \setminus \{0\}$. For each index $1 \leq i \leq n$, define the prime ideal $P_i := (f_i(x, y))$, which $i \neq j$ implies $P_i \neq P_j$ (since their generators are non-associates). Notice that since f is divisible by each f_i , one has $(f) \subseteq (f_i) = P_i$ for all index i .

Now, we claim that $V((f)) = \bigcup_{i=1}^n V(P_i)$:

- Given any prime ideal $Q \in \bigcup_{i=1}^n V(P_i)$, since there exists index i such that $Q \in V(P_i)$, one has $P_i \subseteq Q$; as a result, $(f) \subseteq P_i \subseteq Q$, showing $Q \in V((f))$. Hence, $\bigcup_{i=1}^n V(P_i) \subseteq V((f))$.
- Conversely, for any prime ideal $Q' \in V((f))$, since $\prod_{i=1}^n f_i = f \in (f) \subseteq Q'$, then there exists index i with $f_i \in Q'$ based on the prime property. Hence, $P_i = (f_i) \subseteq Q'$, showing $Q' \in V(P_i)$. So, $V((f)) \subseteq \bigcup_{i=1}^n V(P_i)$.

This provides the desired finite union, or $X = V((f)) = \bigcup_{i=1}^n V(P_i)$.

Proof of (2):

To prove (2), it suffices to show that $i \neq j$ implies $V(P_i) \cap V(P_j)$ is finite (since all arbitrary intersections of $V(P_i)$'s is contained in some $V(P_i) \cap V(P_j)$ for $i \neq j$; if each such pair is finite, with finitely many pairs of (i, j) possible, S must also be finite).

Given $i \neq j$ (where $P_i \neq P_j$), consider the ideal $P_i + P_j$:

- If $P_i + P_j = \mathbb{C}[x, y]$, then $V(P_i) \cap V(P_j) = V(P_i + P_j) = V(\mathbb{C}[x, y]) = \emptyset$, so the intersection is vacuously finite.
- Else if $P_i + P_j \neq \mathbb{C}[x, y]$, consider the ring $\mathbb{C}[x, y]/(P_i + P_j)$: Given any prime ideal $\overline{M} \subsetneq \mathbb{C}[x, y]/(P_i + P_j)$, notice that under the projection $\pi: \mathbb{C}[x, y] \twoheadrightarrow \mathbb{C}[x, y]/(P_i + P_j)$, one has $\pi^{-1}(\overline{M}) \subsetneq \mathbb{C}[x, y]$ being a prime ideal containing $P_i + P_j$, hence strictly containing P_i and P_j . Then, since $\text{height}(P_i) = 1$, then $\text{height}(\pi^{-1}(\overline{M})) > 1$ and ≤ 2 (recall that the Krull dimension of $\mathbb{C}[x, y]$ is 2, and height of a prime ideal is at most the Krull dimension of the ring), hence $\pi^{-1}(\overline{M})$ has height 2, showing it's a maximal ideal. Which, this enforces $\overline{M} \subsetneq \mathbb{C}[x, y]/(P_i + P_j)$ to also be a maximal ideal.

This shows that all prime ideal in $\mathbb{C}[x, y]/(P_i + P_j)$ is maximal (or, the Krull Dimension of $\mathbb{C}[x, y]/(P_i + P_j)$ is 0); moreover, because $\mathbb{C}[x, y]$ is Noetherian, then $\mathbb{C}[x, y]/(P_i + P_j)$ as its quotient is also Noetherian.

Here, recall a commutative algebra fact: A commutative ring A is Artinian $\iff A$ is Noetherian and its Krull Dimension is 0. So, since $\mathbb{C}[x, y]/(P_i + P_j)$ is Noetherian and has Krull Dimension 0, it's Artinian. As a consequence, it must have finite maximal ideal (which are all the prime ideals, since in Artinian ring all prime ideals are maximal), showing that $\text{Spec}(\mathbb{C}[x, y]/(P_i + P_j))$ is finite.

Finally, since one has $V(P_i) \cap V(P_j) = V(P_i + P_j) \cong \text{Spec}(\mathbb{C}[x, y]/(P_i + P_j))$, then the intersection is finite. And, it only contains maximal ideal, since any prime ideal $M \in V(P_i) \cap V(P_j)$ (containing $P_i + P_j$) also strictly contains P_i (which has height 1), so M must have height 2 = $\dim(\mathbb{C}[x, y])$, showing M is maximal.

Since in each case $V(P_i) \cap V(P_j)$ (for $i \neq j$) is finite, and there are finitely many pairs of distinct i, j , then the set S collecting all nontrivial intersections between $V(P_i)$'s is also a finite set. Also, within the proof we've indicated that every such nontrivial intersection must consist of maximal ideals. Hence, S is a finite set of maximal ideals.

Proof of (3):

Now, consider the set $X \setminus S$. Using (1) and (2), we get it's the same as $(\bigcup_{i=1}^n V(P_i)) \setminus S = \bigcup_{i=1}^n (V(P_i) \setminus S)$. However, notice that S contains all the nontrivial intersection of each $V(P_i)$'s, hence one has $(V(P_i) \setminus S) \cap (V(P_j) \setminus S) = \emptyset$ for all $i \neq j$. Therefore, $X \setminus S = \bigsqcup_{i=1}^n (V(P_i) \setminus S)$, and each subset $V(P_i) \setminus S = V(P_i) \cap (X \setminus S)$ is closed in $X \setminus S$ under its subspace topology.

Given that each $P_i \subset \mathbb{C}[x, y]$ is a prime ideal, then $V(P_i) \cong \text{Spec}(\mathbb{C}[x, y]/P_i)$ as topological spaces is irreducible (since $\mathbb{C}[x, y]/P_i$ is an integral domain, its prime spectrum is irreducible, a proposition proven in class).

Which, since the set S collects finitely many maximal ideals (which $M \in S$ a maximal ideal implies $\{M\} = V(M)$ is a closed set), hence S is a finite union of closed points in $\text{Spec}(\mathbb{C}[x, y])$, which is also closed in $\text{Spec}(\mathbb{C}[x, y])$.

As a result, $V(P_i) \setminus S = V(P_i) \setminus (V(P_i) \cap S)$ is open under its own subspace topology. So, if it's nonempty (we'll ignore the empty ones), $V(P_i) \setminus S \subseteq V(P_i)$ is an open subset of an irreducible space, hence is also irreducible.

This concludes that each nonempty $V(P_i) \setminus S$ is irreducible (which is also connected).

Proof of (4):

Finally, with $X \setminus S = \bigsqcup_{i=1}^n (V(P_i) \setminus S)$, each nonempty $(V(P_i) \setminus S) \subseteq (X \setminus S)$ is closed, irreducible, and connected (by (3)). Then, each nonempty $V(P_i) \setminus S$ must be a connected component: Suppose $(V(P_i) \setminus S) \subseteq U$ for some connected subset $U \subseteq (X \setminus S)$. As a result, $U = U \cap (X \setminus S) = \bigsqcup_{i=1}^n (U \cap (V(P_i) \setminus S))$. Since each $V(P_i) \setminus S$ is disjoint from each other, U here is written as a finite disjoint union of closed subsets (under subspace topology of U). Which, by connectedness, with $V(P_i) \setminus S \subseteq U$ being nonempty, the other $j \neq i$ must have $U \cap (V(P_j) \setminus S) = \emptyset$. Hence, $U = U \cap (V(P_i) \setminus S)$, showing that $U \subseteq V(P_i) \setminus S$, or $U = V(P_i) \setminus S$.

Hence, each nonempty $V(P_i) \setminus S$ is indeed a connected component of $X \setminus S$ that's irreducible; then, with $X \setminus S = \bigsqcup_{i=1}^n (V(P_i) \setminus S)$, the nonempty $V(P_i) \setminus S$ must be all the connected components, showing S is a finite set which $X \setminus S$ has all connected component being irreducible.

This is the desired result for the problem. □

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Problem 2. Let $f : X \rightarrow Y$ be a morphism of integral schemes which is a topological homeomorphism. Prove or disprove that f is an isomorphism of schemes.

Solution. We'll disprove the statement by constructing a counterexample.

Consider the field inclusion $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$, which generates the pullback map $\iota^* : \text{Spec}(\mathbb{C}) \rightarrow \text{Spec}(\mathbb{R})$.

First, notice that because $\mathbb{R}, \mathbb{C}$ are both fields, the $\text{Spec}(\mathbb{C}), \text{Spec}(\mathbb{R})$ are both singleton set, hence as continuous maps between topological spaces, ι^* is a topological homeomorphism.

Second, since they're singletons, then the only nonempty open subsets are $\text{Spec}(\mathbb{C})$ and $\text{Spec}(\mathbb{R})$ themselves, so one deduces $\mathcal{O}(\text{Spec}(\mathbb{C})) = \mathbb{C}$ and $\mathcal{O}(\text{Spec}(\mathbb{R})) = \mathbb{R}$, which both are integral domains. Hence, these two spaces have all the nonempty open neighborhoods corresponding to an integral domain using their structure sheaves, showing $\text{Spec}(\mathbb{C})$ and $\text{Spec}(\mathbb{R})$ are integral schemes.

Yet, it's not an isomorphism as schemes: Suppose the contrary that it's an isomorphism of schemes, there exists another morphism of integral schemes $j : \text{Spec}(\mathbb{R}) \rightarrow \text{Spec}(\mathbb{C})$, such that it's not only an inverse of ι^* , but its induced ring homomorphism $j^* : \mathcal{O}(\text{Spec}(\mathbb{C})) \rightarrow \mathcal{O}(\text{Spec}(\mathbb{R}))$ is the inverse of the ring homomorphism $(\iota^*)^* : \mathcal{O}(\text{Spec}(\mathbb{R})) \rightarrow \mathcal{O}(\text{Spec}(\mathbb{C}))$. However, the induced ring homomorphism of ι^* , or $(\iota^*)^*$, is precisely the inclusion $\iota : \mathbb{R} \hookrightarrow \mathbb{C}$; since $j^* : \mathbb{C} \rightarrow \mathbb{R}$ is its inverse, they imply that $\mathbb{R}$ and $\mathbb{C}$ are isomorphic fields.

This is a contradiction, because consider $i \in \mathbb{C}$, its multiplicative order is 4 (since $i^4 = 1$, and any positive integer $n < 4$ has $i^n \neq 1$); however, there's no element in $\mathbb{R}$ with multiplicative order of 4 (since if $x \in \mathbb{R}$ satisfies $x^4 = 1$, then $x^4 - 1 = (x - 1)(x + 1)(x^2 + 1) = 0$; with $x^2 + 1 = 0$ having no solution in $\mathbb{R}$, then $x = \pm 1$, which it either has multiplicative order of 1 or 2, not 4). hence, $\mathbb{R}, \mathbb{C}$ can't be isomorphic fields, leading to a contradiction.

So, since we derive a contradiction, $\iota^* : \text{Spec}(\mathbb{C}) \rightarrow \text{Spec}(\mathbb{R})$ can't be an isomorphism of schemes, despite it's a topological homeomorphism. □

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Problem 3. Let k be a countable field and let $f : X \rightarrow \text{Spec}(k)$ be a finite type morphism of schemes. Prove or disprove that X has at most countably many points.

Solution. We'll prove that X has at most countably many points.

First, since $\text{Spec}(k)$ is an affine scheme, and k is a field (which is Noetherian), it's a locally Noetherian Scheme. Moreover, $\text{Spec}(k)$ is a singleton, hence quasi-compact, so $\text{Spec}(k)$ is a Noetherian scheme. Which, with $f : X \rightarrow \text{Spec}(k)$ being a finite type morphism and $\text{Spec}(k)$ being a Noetherian scheme, X must be Noetherian also (based on a theorem proven in class).

Then, using the definition of finite type morphism, there exists an affine cover $\{U_i\}$ of $\text{Spec}(k)$, each U_i corresponds to finite affine open subset $\{U_{ij}\}$ in X , such that $f^{-1}(U_i) = \bigcup_j U_{ij}$, and the induced ring homomorphism $\mathcal{O}(U_i) \rightarrow \mathcal{O}(U_{ij})$ is a finite type ring homomorphism.

Notice that since $\text{Spec}(k)$ is compact, the list $\{U_i\}$ can be chosen to be finite (moreover, with $\text{Spec}(k)$ being singleton, it can be chosen to be a single set U). Which, since U covers $\text{Spec}(k)$, one has $U = \text{Spec}(k)$, deducing that $\mathcal{O}(U) = \mathcal{O}(\text{Spec}(k)) = k$. So, given that $f^{-1}(U) = f^{-1}(\text{Spec}(k)) = X$, one has $X = \bigcup_j U_j$ for finitely many affine open subsets U_j , and each induced ring homomorphism $\mathcal{O}(U) \rightarrow \mathcal{O}(U_j)$ is of finite type.

With $\mathcal{O}(U) = k$, then the given ring homomorphism is of finite type implies there exists a surjective k -algebra homomorphism $k[x_1, \dots, x_n] \twoheadrightarrow \mathcal{O}(U_j)$, where n is a finite number (in other words, each $\mathcal{O}(U_j)$ is a quotient of finite-variable polynomial rings over k).

Now, we claim that each U_j is in fact countable: Based on the above fact, with $\mathcal{O}(U_j)$ being a quotient of $k[x_1, \dots, x_n]$ for some finite n , then $U_j = \text{Spec}(\mathcal{O}(U_j))$ can be interpreted as a subset of $\text{Spec}(k[x_1, \dots, x_n])$ (since given a commutative ring A with an ideal $I \subset A$, $\{\text{ideal } J \subseteq A/I\}$ and $\{\text{ideal } K \subset A \mid I \subseteq K\}$ are of bijective correspondence, while preserving prime ideals). Hence, to prove U_j is countable, it suffices to prove that $\text{Spec}(k[x_1, \dots, x_n])$ is countable (given that k is a countable field). We'll prove it as follow:

(1) **$k[x_1, \dots, x_n]$ is countable, given that k is countable:**

Here we'll approach by induction on the number of variables n . For $n = 1$, recall that every polynomial $f(x) \in k[x]$ is of bounded degree, hence there exists $i \in \mathbb{N}$, such that $f(x) \in \bigoplus_{n=0}^i kx^n$, which is isomorphic to $\bigoplus_{n=0}^i k$ as abelian group. Because finite product of countable sets is countable, and finite product coincides with direct sum for abelian groups, one has $\bigoplus_{n=0}^i kx^n$ being countable. As a result, $k[x] = \bigcup_{i=0}^{\infty} \left(\bigoplus_{n=0}^i kx^n \right)$ is a countable union of countable sets, hence countable.

Now, suppose for given $n \in \mathbb{N}$, all $j < n$ satisfies $k[x_1, \dots, x_j]$ being countable, then for $k[x_1, \dots, x_n]$, interpret as the polynomial ring $(k[x_1, \dots, x_{n-1}])[x_n]$. Because $k[x_1, \dots, x_{n-1}]$ is countable by induction hypothesis, apply the proof for the case $n = 1$, we obtained that $k[x_1, \dots, x_n]$ (as a single-variable polynomial ring over $k[x_1, \dots, x_{n-1}]$) is also countable. This finishes the induction.

(2) **There are at most countable ideals for $k[x_1, \dots, x_n]$:**

Based on Hilbert's Basis Theorem, since k is a field (in particular a Noetherian Ring), $k[x_1, \dots, x_n]$ is a Noetherian Ring. Hence, all ideals of $k[x_1, \dots, x_n]$ are finitely generated.

Now, for each ideal I , fix a specific finite set of generators (say $I = (f_1, \dots, f_i)$), then let $S = \{\text{all finite subset of } k[x_1, \dots, x_n]\}$, there is a clear inclusion from $\Sigma = \{\text{ideals } I \subseteq k[x_1, \dots, x_n]\}$ to S , by $I \mapsto \{f_1, \dots, f_i\}$ (the fixed finite set of generators).

Finally, because $k[x_1, \dots, x_n]$ is countable, let $(f_j)_{j=0}^{\infty}$ be an ordering of $k[x_1, \dots, x_n]$ using natural numbers. Then, for any finite subset $T \subset k[x_1, \dots, x_n]$, it contains some largest index, say $j_T \in \mathbb{N}$, in particular $T \in \mathcal{P}(\{f_0, \dots, f_{j_T}\})$ (the power set of $\{f_0, \dots, f_{j_T}\}$). So as a set, one has

$S \subseteq \bigcup_{j=0}^{\infty} \mathcal{P}(\{f_0, \dots, f_j\})$. Because the index is countable, and each $\mathcal{P}(\{f_0, \dots, f_j\})$ is finite, then the union $\bigcup_{j=0}^{\infty} \mathcal{P}(\{f_0, \dots, f_j\})$ is countable, which its subset S is also countable.

As a result, because the set of ideals Σ can be identified as a subset of S , Σ is also countable, showing $k[x_1, \dots, x_n]$ has countable number of ideals. In particular, $\text{Spec}(k[x_1, \dots, x_n])$ is also countable (since it collects all prime ideals of $k[x_1, \dots, x_n]$).

Finally, since each $\text{Spec}(k[x_1, \dots, x_n])$ is countable, because $\mathcal{O}(U_j)$ is a quotient of $k[x_1, \dots, x_n]$ (for some positive integer n), say $\mathcal{O}(U_j) = k[x_1, \dots, x_n]/I$, then $\text{Spec}(\mathcal{O}(U_j)) = U_j$ is also countable (since $U_j = \text{Spec}(\mathcal{O}(U_j)) = \text{Spec}(k[x_1, \dots, x_n]/I) \cong V(I)$ as a subspace of $\text{Spec}(k[x_1, \dots, x_n])$). Then, because $X = \bigcup_j U_j$, with the index being finite, X itself must be countable as set, hence has at most countably many points. □

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Problem 4. Let X be a scheme which is covered by two affine open subschemes. Prove or disprove that X is affine.

Solution. Here, we'll prove that X is not necessarily affine, by constructing a counterexample.

Consider the two schemes $Y = Z = \mathbb{A}_{\mathbb{C}}^1 = \text{Spec}(\mathbb{C}[t])$, and consider $U = \mathbb{A}_{\mathbb{C}}^1 \setminus \{(t)\}$ (where (t) corresponds to the maximal ideal for origin $0 \in \mathbb{C}$). Then, take the identity map $f = \text{id}_U : U \xrightarrow{\sim} U$, the inclusion $\iota_U : U \hookrightarrow Y = Z$, and consider $X = Y \sqcup_f Z$ as the gluing of schemes:

$$\begin{array}{ccc} U & \xrightarrow{\iota_U} & Y \\ \downarrow \iota_U & & \downarrow \iota_Y \\ Z & \xrightarrow{\iota_Z} & X = Y \sqcup_f Z \end{array}$$

Which, X is clearly covered by Y and Z (by the image of them under the canonical inclusion maps), which both open subsets are affine. Yet, we'll prove that X itself is not affine via contradiction:

Suppose the contrary that X is an affine scheme. If consider the evaluation of the structure sheaves, one induce the following morphisms:

$$\begin{array}{ccc} \mathcal{O}(U) & \xleftarrow{\iota_U^*} & \mathcal{O}(Y) \\ \uparrow \iota_U^* & & \uparrow \iota_Y^* \\ \mathcal{O}(Z) & \xleftarrow{\iota_Z^*} & \mathcal{O}(X) \end{array}$$

Where, $\mathcal{O}(X)$ together with the ring homomorphisms ι_Y^*, ι_Z^* form a fibre product of the two ring homomorphisms $\iota_U^* : \mathcal{O}(Y) = \mathcal{O}(Z) \rightarrow \mathcal{O}(U)$.

Notice that since $U = \mathbb{A}_{\mathbb{C}}^1 \setminus \{(t)\}$, it denotes all the prime ideals in $\mathbb{C}[t]$, which is of the form $(t - a)$ for some $a \neq 0$ (by the fact that $\mathbb{C}$ is algebraically closed, and $\mathbb{C}[t]$ is a PID). So, since if $a \neq 0$ then $t \notin (t - a)$ (because a is not a solution of t , so $t \notin (t - a)$), and t is irreducible in $\mathbb{C}[t]$, showing (t) is the only prime (hence maximal) ideal containing t . This proves that $U = D(t)$, the open set consists of all prime ideals not containing $t \in \mathbb{C}[t]$.

As a result, $\mathcal{O}(U) = \mathcal{O}(D(t)) = \mathbb{C}[t]_t \cong (\mathbb{C}[t])[x]/(tx - 1)$; and, with $Y = Z = \mathbb{A}_{\mathbb{C}}^1$, $\mathcal{O}(Y) = \mathcal{O}(Z) = \mathcal{O}(\text{Spec}(\mathbb{C}[t])) = \mathbb{C}[t]$. This shows that $\iota_U^* : \mathbb{C}[t] \rightarrow \mathbb{C}[t]_t$ is given by composing the inclusion $\mathbb{C}[t] \hookrightarrow (\mathbb{C}[t])[x]$ and projection $(\mathbb{C}[t])[x] \twoheadrightarrow (\mathbb{C}[t])[x]/(tx - 1)$.

• **Note:** For later purpose, we'll prove that $\iota_U^* : \mathbb{C}[t] \rightarrow (\mathbb{C}[t])[x]/(tx - 1)$ is injective:

If $f \in \mathbb{C}[t] \hookrightarrow (\mathbb{C}[t])[x]$ has $f \in (tx - 1)$, notice that as a polynomial ring over $\mathbb{C}[t]$, one has $\deg_x(f) = 0$, if $f = g \cdot (tx - 1)$ for some $g \in (\mathbb{C}[t])[x]$, then $f = 0$ (since if $f \neq 0$, then $0 = \deg_x(f) = \deg_x(g \cdot (tx - 1)) \geq \deg_x(tx - 1) = 1$, which is a contradiction). Therefore, for $f \in \mathbb{C}[t] \hookrightarrow (\mathbb{C}[t])[x]$, one has $\bar{f} = 0 \in (\mathbb{C}[t])[x]/(tx - 1)$ iff $f \in (tx - 1)$ iff $f = 0$, showing the map $\iota_U^* : \mathbb{C}[t] \rightarrow (\mathbb{C}[t])[x]/(tx - 1)$ is injective.

However, since $\iota_U^* : \mathcal{O}(Y) = \mathcal{O}(Z) \rightarrow \mathcal{O}(U)$ are just two identical maps (since they're induced by the same morphism of schemes), then their fibre product is in fact $\mathcal{O}(Y) = \mathcal{O}(Z)$ together with identity, so $\mathcal{O}(X) \cong \mathcal{O}(Y) = \mathcal{O}(Z) = \mathbb{C}[t]$, and $\iota_Y^* = \iota_Z^* = \text{id}_{\mathbb{C}[t]}$.

(Note: Take $\mathcal{O}(X) = \mathcal{O}(Y) = \mathcal{O}(Z) = \mathbb{C}[t]$ and $\iota_Y^* = \iota_Z^* = \text{id}_{\mathbb{C}[t]}$ definitely satisfies the fibre square, because the two maps taking fibre product are identical; Now, given any commutative ring B together with ring homomorphisms $i_Y : B \rightarrow \mathcal{O}(Y)$ and $i_Z : B \rightarrow \mathcal{O}(Z)$ such that $\iota_U^* \circ i_Y = \iota_U^* \circ i_Z$, i.e. B with i_Y, i_Z satisfies the fibre squares; then, because ι_U^* is proven to be injective, then $\iota_U^* \circ i_Y = \iota_U^* \circ i_Z$ enforces $i_Y = i_Z$ as set maps. So, $i_Y = i_Z : B \rightarrow \mathcal{O}(Y) = \mathcal{O}(Z) = \mathbb{C}[t]$ is the unique map satisfying $\text{id}_{\mathbb{C}[t]} \circ i_Y = i_Y$, and $\text{id}_{\mathbb{C}[t]} \circ i_Z = i_Z$).

Finally, since X is assumed to be affine, and $\mathcal{O}(X) = \mathbb{C}[t]$ based on the above observation, one has $X = \text{Spec}(\mathbb{C}[t]) = \mathbb{A}_{\mathbb{C}}^1$. Yet, notice that this creates a contradiction:

Recall that morphisms between affine schemes must be generated by unique ring homomorphism between their corresponding rings. Which, $\iota_Y : Y \rightarrow X$ is induced by $\iota_Y^* = \text{id}_{\mathbb{C}[t]} : \mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ (or $\iota_Y = (\iota_Y^*)^*$); however, notice that taking the induced morphism of affine schemes of $\text{id}_{\mathbb{C}[t]}$, we get $(\text{id}_{\mathbb{C}[t]})^* : Y \rightarrow X$, which is precisely the identity on $X = \text{Spec}(\mathcal{O}(X)) = Y = \text{Spec}(\mathcal{O}(Y)) = \text{Spec}(\mathbb{C}[t])$, this contradicts the fact that $\iota_Y : Y \rightarrow X$ is not surjective (since when taking the adjunction space, the ideal $z = (t) \in Z$ is not part of $U = \mathbb{A}_{\mathbb{C}}^1 \setminus \{(t)\}$, hence when taking $X = Y \sqcup_f Z$, $z \notin \text{im}(\iota_Y)$).

Since we reach a contradiction, then our assumption is false, which X cannot be affine. Therefore, it's a desired counterexample for this statement. □

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Problem 5. Let $A = \mathbb{R}[X_1, X_2, X_3]/(X_2 - X_1^2, X_3 - X_1^3)$ and $X = \text{Spec}(A)$. Let $f(X_1, X_2, X_3) = a_1X_1 + a_2X_2 + a_3X_3 + a_4$, where $a_i \in \mathbb{R}$ and $Y = \text{Spec}(\mathbb{R}[X_1, X_2, X_3]/(f))$. Is the set $X \cap Y$ finite? If yes, what is a finite upper bound on its cardinality? Explain your answer with complete argument.

Solution. We'll prove that $X \cap Y$ is not necessarily finite, by showing it's infinite for $f = 0$.

Here we denote $I = (X_2 - X_1^2, X_3 - X_1^3)$, and utilized the fact that $X = \text{Spec}(\mathbb{R}[X_1, X_2, X_3]/I) \cong V(I)$, and $Y = \text{Spec}(\mathbb{R}[X_1, X_2, X_3]/(f)) \cong V((f))$ as subspaces in $\text{Spec}(\mathbb{R}[X_1, X_2, X_3])$. Which, the intersection $X \cap Y$ is precisely all the prime ideals $P \subset \mathbb{R}[X_1, X_2, X_3]$ that contains both I and (f) .

Here's an order of the proof:

- (1) Show that $\mathbb{R}[X_1, X_2, X_3]/I \cong \mathbb{R}[t]$.
- (2) Show that $X \cap Y$ has a one-to-one correspondance to a specific closed subset of $\text{Spec}(\mathbb{R}[t])$.
- (3) Deduce $X \cap Y$ is infinite for the case $f = 0$.

Proof of (1):

Consider the evaluation map $\varphi : \mathbb{R}[X_1, X_2, X_3] \twoheadrightarrow \mathbb{R}[t]$ by $\varphi(f(X_1, X_2, X_3)) = f(t, t^2, t^3)$. If consider $X_2 - X_1^2, X_3 - X_1^3$ (the generators of I), one has the following:

$$\varphi(X_2 - X_1^2) = t^2 - (t)^2 = 0, \quad \varphi(X_3 - X_1^3) = t^3 - (t)^3 = 0 \quad (5.1)$$

This shows that $X_2 - X_1^2, X_3 - X_1^3 \in \ker(\varphi)$, so $I \subseteq \ker(\varphi)$.

Now, we argue that $I = \ker(\varphi)$: Given any monomial $X_1^{n_1}X_2^{n_2}X_3^{n_3}$, then one has the following equality:

$$\begin{aligned} X_1^{n_1}X_2^{n_2}X_3^{n_3} &= (X_1^{n_1}X_2^{n_2}X_3^{n_3} - X_1^{n_1+2n_2}X_3^{n_3}) + X_1^{n_1+2n_2}X_3^{n_3} \\ &= X_1^{n_1}X_3^{n_3}(X_2^{n_2} - X_1^{2n_2}) + (X_1^{n_1+2n_2}X_3^{n_3} - X_1^{n_1+2n_2+3n_3}) + X_1^{n_1+2n_2+3n_3} \\ &= X_1^{n_1}X_3^{n_3} \left(\sum_{i=0}^{n_2-1} X_2^i (X_1^2)^{n_2-i-1} \right) (X_2 - X_1^2) + X_1^{n_1+2n_2} (X_3^{n_3} - X_1^{3n_3}) + X_1^{n_1+2n_2+3n_3} \\ &= X_1^{n_1}X_3^{n_3} \left(\sum_{i=0}^{n_2-1} X_2^i (X_1^2)^{n_2-i-1} \right) (X_2 - X_1^2) + X_1^{n_1+2n_2} \left(\sum_{j=0}^{n_3-1} X_3^j (X_1^3)^{n_3-j-1} \right) (X_3 - X_1^3) + X_1^{n_1+2n_2+3n_3} \end{aligned} \quad (5.2)$$

Hence, all monomials can be written as $X_1^{n_1}X_2^{n_2}X_3^{n_3} = f \cdot (X_2 - X_1^2) + g \cdot (X_3 - X_1^3) + X_1^k$ for some $f, g \in \mathbb{R}[X_1, X_2, X_3]$, and $k \in \mathbb{N}$. As a result, all polynomials $h(X_1, X_2, X_3) = f \cdot (X_2 - X_1^2) + g \cdot (X_3 - X_1^3) + h_1(X_1)$ for some $f, g \in \mathbb{R}[X_1, X_2, X_3]$, and $h_1(t) \in \mathbb{R}[t]$.

So, one has $h \in \ker(\varphi)$ iff $\varphi(h) = h_1(t) = 0$ (since $X_2 - X_1^2, X_3 - X_1^3$ are evaluated to 0), which happens iff $h(X_1, X_2, X_3) = f \cdot (X_2 - X_1^2) + g \cdot (X_3 - X_1^3) \in I$. This concludes that $I = \ker(\varphi)$. As a result, $\mathbb{R}[X_1, X_2, X_3]/I \cong \mathbb{R}[t]$.

Proof of (2):

Since $(X \cap Y) \subset X = V(I) \cong \text{Spec}(\mathbb{R}[X_1, X_2, X_3]/I) \cong \text{Spec}(\mathbb{R}[t])$, it's naturally identified as a subset of $\text{Spec}(\mathbb{R}[t])$ (the identification $(X \cap Y) \rightarrow \text{Spec}(\mathbb{R}[t])$ is by $P \mapsto \varphi(P)$, where P is the evaluation map defined in (1)). Notation wise we'll still use $X \cap Y$ (as subset in $\text{Spec}(\mathbb{R}[t])$).

- Notice that all prime ideals $P \in (X \cap Y)$ also contains $f = a_1X_1 + a_2X_2 + a_3X_3 + a_4$, hence it satisfies $\varphi(f) = a_1t + a_2t^2 + a_3t^3 + a_4 \in \varphi(P)$, so $(X \cap Y) \subseteq V((a_1t + a_2t^2 + a_3t^3 + a_4))$ in $\text{Spec}(\mathbb{R}[t])$.
- Conversely, given prime ideal $\bar{P} \in V((a_1t + a_2t^2 + a_3t^3 + a_4))$, then under the preimage of evaluation map, one has $f = a_1X_1 + a_2X_2 + a_3X_3 + a_4 \in \varphi^{-1}(a_1t + a_2t^2 + a_3t^3 + a_4) \subset \varphi^{-1}(\bar{P})$, showing that $\varphi^{-1}(\bar{P})$ is a prime ideal containing both I and (f) , which $\bar{P}$ is in $X \cap Y$ (when identified in $\text{Spec}(\mathbb{R}[t])$).

Therefore, set wise one can conclude that $X \cap Y = V((a_1t + a_2t^2 + a_3t^3 + a_4))$.

Proof of (3):

Part (2) provides that $X \cap Y = V((a_1t + a_2t^2 + a_3t^3 + a_4))$ as set bijection. However, if take $f = a_1X_1 + a_2X_2 + a_3X_3 + a_4 = 0$, this ideal becomes (0) , hence $X \cap Y = V((0))$ in $\text{Spec}(\mathbb{R}[t])$, which is precisely the whole space. However, $\text{Spec}(\mathbb{R}[t])$ is infinite (since $t - a \in \mathbb{R}[t]$ is irreducible for all $a \in \mathbb{R}$, so each $(t - a)$ is a distinct prime ideal in $\text{Spec}(\mathbb{R}[t])$). Hence, $f = 0$ is a case where $X \cap Y$ is infinite, showing $X \cap Y$ is not necessarily finite. This completes the proof. □